MIL-I- 45691(Ord)
8 December 1960
SUPERSEDING
FA(FC)-PD-127
20 April 1959

# MILITARY SPECIFICATION

## INVERTER, POWER, STATIC: 10512741

### 1. SCOPE

1.1 This specification covers one type of 400 cycle, 115 volt, static inverter intended for use with an ~~electric~~ electronic analog ballistics computer used in tank fire control systems.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |
| MIL-F-14619 | Fire Control Parts Equipment and Tools; Packaging of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |

DRAWINGS

Ordnance Corps

| | |
|---|---|
| C10512741 | Inverter Power Static: 10512741 |

FSC 1220

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Production model.- At the beginning of production one inverter representative of production shall be made available to the Government to determine if the inverter meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The inverter shall be manufactured in accordance with Drawing C10412741, "Inverter, Power, Static:10512741" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Protective finished surface.
(c) Special accuracy.
(d) Finish painting and protective.
(e) Electrical.
(f) Workmanship.

3.5 Environmental.

3.5.1 Operating temperatures.- The inverter shall be operated for a period of at least eight hours in a thermally stabilized ambient temperature area of plus 150 degrees Fahrenheit. At the end of the eight hour operational period, the inverter while still operating at the 150 degrees Fahrenheit (plus increased temperature resulting from continuous inverter operation) shall meet the requirements of 3.6.4. Subsequent to meeting the high temperature requirement the inverter shall then be thermally stabilized for a minimum of one hour at minus 40 degrees Fahrenheit and shall be operated to meet the requirements of 3.6.4 while at that temperature. An external heat sink shall not be utilized for this requirement.



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3.5.1.1 Durability.- With the inverter at an ambient temperature of plus 125 degrees plus or minus 5 degrees Fahrenheit the inverter energized with 29 volts plus or minus 0.5 volt dc (see 6.5) and with a load of 80 volt ampers and a power factor of between 0.8 leading and 0.4 lagging, the inverter shall operate continuously for a period of 8 hours after which it shall meet the requirements of 3.6.4.2, 3.6.4.3 and 3.6.4.4 when checked at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

3.5.2 Storage temperatures.- The inverter shall show no evidence of physical failure and shall meet all performance requirements of this specification at standard ambient temperature after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.5.3 Vibration.- The inverter shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.060 inch total excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and returned to 10 cycles and shall be traversed in approximately one minute. This motion shall be applied for a period of 10 minutes in each of 3 mutually perpendicular direction (total of 30 minutes). Subsequent to vibration the inverter shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.4 Humidity.- The inverter with cover in place shall be exposed to one complete humidity cycle as shown on Figure 1. Immediately following completion of the cycle, the cover shall be removed and the electronic components shall not be damaged or be adversely affected and the unit shall meet all operational requirements of 3.6.4.

3.5.5 Water resistance.- The inverter shall meet the requirements of Procedure I of the rain test of Specification MIL-F-13926.

3.6 Performance.

3.6.1 Continuity.- The electrical circuits shall be tested for continuity and shall conform to the drawing requirements.

3.6.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the inverter are subjected to an alternating current (ac) root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than five seconds.



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3.6.3 Insulation resistance.- Unless otherwise specified on applicable, drawings, all insulation between ungrounded circuits and ground, and between isolated circuits shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.6.4 Operation.- The inverter shall meet the following operational requirements when input voltages, with at least 10 amperes capacity, within the range of 18 to 30 volts dc are applied (see 6.5).

3.6.4.1 Input current drawn.- The input current drawn shall not exceed 5 amperes.

3.6.4.2 Output voltage.- The output voltage shall be 115 volts rms plus or minus 5 percent.

3.6.4.3 Output volt amperes.- The normal output volt amperes shall be 80.

3.6.4.4 Operating frequency.- The operating frequency shall be 400 cycles plus or minus 2 percent.

3.6.4.5 Distortion.- Total harmonic distortion shall be less than 4 percent.

3.6.4.6 Power factor.- The power factor (P.F.) with a load of 100 volt amperes shall be between 0.8 lagging to 0.6 leading.

3.7 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.



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4.4 Production model.- Unless otherwise specified in the contract, one inverter shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production model failure.- Should the production model fail to pass the prescribed examinations and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.2 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelsim micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the inverter. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.5.2.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.



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TABLE I - SAMPLING PLAN

| Lot Size Inclusive | Sample Size | Accept Lot | Reject Lot |
|---|---|---|---|
| 0 - 25 | Use 100% Inspection | | |
| 26 - 35 | 20 | 0 | 1 |
| 36 - 50 | 25 | 0 | 1 |
| 51 - 150 | 30 | 0 | 1 |
| 151 - 300 | 30 | 1 | 2 |
| 301 - over | 60 | 2 | 3 |

4.5.2.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specificatinns. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitable equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

4.7.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirements, and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.



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## TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

| MAJOR: AQL 4.0% | Req't Par. | QA. Par. | Test Method |
|---|---|---|---|
| 101. Operating temperatures | 3.5.1 | 4.9.1.1 | Standard Measuring Equipment (SME) |
| 102. Physical failure, post vibration | 3.5.3 | 4.9.1.4 | Device, Special Testing (DST) |

MINOR: NONE DEFINED

## TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

| MAJOR: AQL 1.5% | Req't Par. | QA. Par. | Test Method |
|---|---|---|---|
| 101. Continuity | 3.6.1 | 4.9.2.1 | Standard Measuring Equipment (SME) |
| 102. Dielectric strength | 3.6.2 | 4.9.2.2 | SME |
| 103. Insulation resistance | 3.6.3 | 4.9.2.3 | SME |
| 104. Input current | 3.6.4.1 | 4.9.2.4.1 | SME |
| 105. Output voltage | 3.6.4.2 | 4.9.2.4.2 | SME |
| 106. Output current | 3.6.4.3 | 4.9.2.4.3 | SME |
| 107. Operating frequency | 3.6.4.4 | 4.9.2.4.4 | SME |
| 108. Distortion | 3.6.4.5 | 4.9.2.4.5 | SME |
| 109. Power factor | 3.6.4.6 | 4.9.2.4.6 | SME |
| 110. Identification marking | 3.7 | 4.9.3 | Visual |

MINOR: NONE DEFINED

4.8 Inspection equipment.



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4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor shall also maintain records of inspection equipment surveillance and calibration, and shall make these records available to the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements shall be capable of measuring these values within an accuracy of 10 percent of the specified tolerance, except in those cases specified otherwise. In cases where the 10 percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2 Accuracy requirements for special testing device.

4.8.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion at a uniform varied frequency between the limits of 10 and 55 cycles per second. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion), and shall be equipped with a medium capable of holding the unit securely to the vibrating machine so that motion is applied to the unit in each of the 3 mutually perpendicular planes as required. The tolerances for amplitude and frequency contained in Specification MIL-F-13926 shall apply.

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Operating temperatures.- This test shall be conducted in accordance with procedure I of Specification MIL-F-13926. When the inverter is thermally stabilized at conditions "C" and "G" the inverter shall be energized and shall be subjected to the test specified in 4.9.1.2.2.



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4.9.1.2 Durability.

4.9.1.2.1 Sampling.- One of the first five inverters produced, and one (1) inverter from each lot of fifty (50) produced thereafter shall be subjected to the durability test. The failure of the inverter to meet the requirements shall be cause for immediate suspension of Government acceptance until necessary corrective measures have been taken to the satisfaction of the contracting officer.

4.9.1.2.2 Testing.- The inverter shall be placed in a chamber and thermally stabilized at the specified temperatures. Upon thermal saturation at plus 150 degrees Fahrenheit the inverter shall then operate for a minimum of eight hours with the specified load applied, and with the initial temperature of plus 150 degrees Fahrenheit being constantly applied the inverter shall meet the requirements specified. The additional temperature created by inverter operation during the eight hour period will cause the temperature of the inverter and the temperature area to rise above the constantly applied temperature of 150 degrees Fahrenheit. Upon thermal saturation at minus forty degrees Fahrenheit the inverter shall operate for a minimum of one hour with the specified load applied and the inverter shall meet the requirements specified. At the completion of the high and low temperature test the inverter shall be returned to standard ambient temperature and subjected to the test specified in 4.9.2.4.2, 4.9.2.4.4 and 4.9.2.4.6 and shall meet the requirements specified.

4.9.1.3 Storage temperatures.- This test shall be conducted in accordance with Procedure II, conditions "A" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the unit shall be inspected for evidence of physical failure and then subjected to tests specified in 4.9.2.1 to 4.9.2.4.7 inclusive, and shall meet the requirements specified. These tests shall be performed on the production model only.

4.9.1.4 Vibration.- This test shall be conducted in accordance with Method 201 of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the unit shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all the tests specified in this specification.



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4.9.1.5 Humidity.- The inverter unit shall be placed in the test chamber in its normal operating position and exposed to one complete humidity condensation cycle as specified. Distilled or demineralized water having a $p^H$ value between 6.5 and 7.5 at 77 degrees Fahrenheit shall be used to obtain the desired humidity. The unit under test shall be placed no closer than 6 inches from the sides of the chamber, and shall not be subject to radiant heat. Upon completion of the humidity exposure, the unit shall show no evidence of damage, and shall then be subjected to the tests in 4.9.2.4 to 4.9.2.4.6 inclusive, and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.1.6 Water resistance.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926. The inverter unit shall show no evidence of water penetration as a result of this test. The test shall be performed on the production model only.

4.9.2 Performance.

4.9.2.1 Continuity.- A continuity check of the unit circuits shall be made by the use of standard measuring equipment to determine conformance with the requirements specified.

4.9.2.2 Dielectric strength.- The inverter unit shall be tested in accordance with Method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.9.2.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the unit shall be given the insulation resistance test in accordance with Method 302, test condition "B", of Standard MIL-STD-202, and shall meet the requirements specified.

4.9.2.4 Operation.- The following tests shall be conducted with the input voltage(dc)to the inverter established as specified.

4.9.2.4.1 Input current.-The measured input current drawn shall not exceed the ampere value specified. This measurement shall be made by the use of standard measuring equipment.

4.9.2.4.2 Output voltage.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured RMS output voltage shall be within the tolerance specified. This measurement shall be made by use of standard measuring equipment.



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4.9.2.4.3 Output current.- The measured normal output current shall be the number of volt amperes specified. This measurement shall be made by use of standard measuring equipment.

4.9.2.4.4 Operating frequency.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured operating frequency shall be within the tolerance specified. The measurement shall be made by use of standard measuring equipment.

4.9.2.4.5 Distortion.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured total harmonic distortion shall not exceed the tolerance specified. This measurement shall be made by use of standard measuring equipment.

4.9.2.4.6 Power factor.- The measured power factor, with a load of 100 volt amperes, shall not exceed the tolerance specified. This measurement shall be made by use of standard measuring equipment.

4.9.2.4.7 Input voltages.- While the input supply voltages are varied between 18 and 30 volts(dc) the test specified in 4.9.2.4.2, 4.9.2.4.4 and 4.9.2.4.5 shall be conducted. This test shall be conducted on the production model only.

4.9.3 Identification marking.- Markings shall be visually inspected for conformance to Standard MIL-STD-130.

4.9.4 Final acceptance.- Final acceptance of a lot of inverter units shall be withheld by the Government inspector until the sample units, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The inverter shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-F-14619-P9547. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use.- This static inverter is primary intended for use with the Electronic Analog Ballistic Computer XM16C and is designed to draw from 24 volts storage batteries and invert this voltage to 115 volts dc, 400 cycles.



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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of protection.
(c) Applicable stock number.
(d) Provision for production model including date of delivery (see 3.1).

6.3 Design information.- Information regarding specific design not covered or referenced by this specfication may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Device, special testing (DST).- Special testing devices are designed to check function of an assembled unit.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.5 Input voltage.- Caution, at no time shall an input voltage or an instantaneous input voltage surge, be applied to the input of the inverter in excess of 30 volts dc. The application of an excessive voltage gradually or instantaneous will result in permanent damage to the transistors and render the inverter unserviceable.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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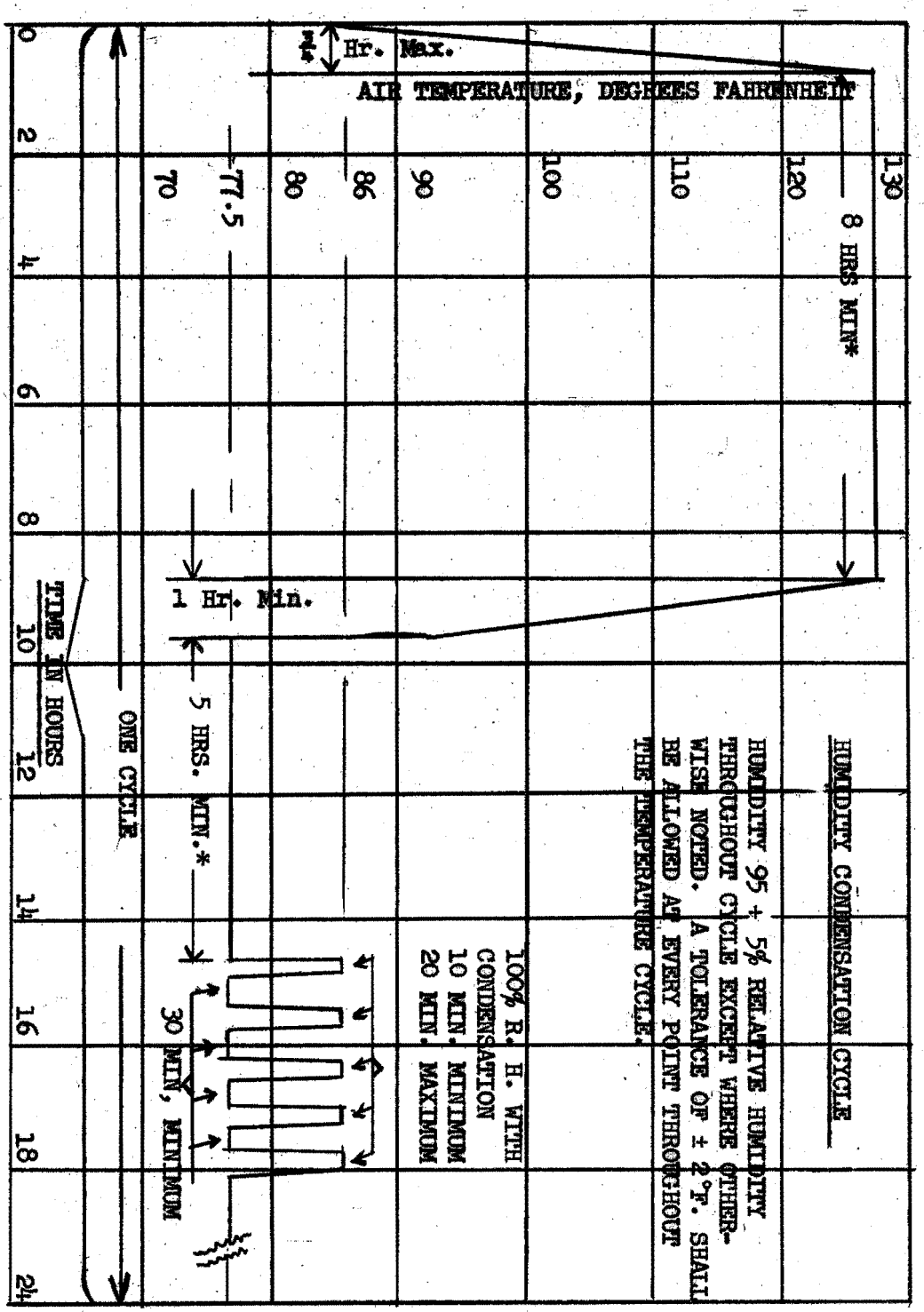



FIGURE I

* OR UNTIL EQUIPMENT STABILIZATION (WHICHEVER IS LONGER)



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